

Elementary Number Theory

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.33.1 and Mathlib v4.33.1 (revision [0df444a360ea](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.
- **Prerequisites.** Starting assumptions and accepted external results are specific to this textbook.

Chapter 1

Preliminaries

1.1 Mathematical Induction

1.2 The Binomial Theorem

Chapter 2

Divisibility Theory in the Integers

2.1 Early Number Theory

2.2 The Division Algorithm

Theorem 2.2.1.

Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}^+$. There is a unique pair $(q, r) \in \mathbb{Z} \times \mathbb{N}$ such that

$$a = qb + r, \quad r < b.$$

We call q the **quotient** and r the **remainder**.

```
theorem existsUnique_quotient_remainder (a : ℤ) (b : ℕ+) :  
  ∃! qr : ℤ × ℕ, a = qr.1 * b + qr.2 ∧ qr.2 < b
```

Proof.

Since $b \geq 1$,

$$a - (-|a|)b = a + |a|b \geq a + |a| \geq 0.$$

```
classical  
have hb : (1 : ℤ) ≤ (b : ℤ) := by exact_mod_cast b.pos  
have hnonneg : 0 ≤ a - (-|a|) * (b : ℤ) := by  
  have ha : -|a| ≤ a := neg_abs_le a  
  have habs : 0 ≤ |a| := abs_nonneg a  
  nlinarith
```

Define

$$S := \{n \in \mathbb{N} : a - qb = n \text{ for some } q \in \mathbb{Z}\}.$$

By the preceding inequality, $S \neq \emptyset$. Let $r = \min S$ and choose $q \in \mathbb{Z}$ with $a - qb = r$.

```

let h :  $\exists n : \mathbb{N}, \exists q : \mathbb{Z}, a - q * b = (n : \mathbb{Z}) :=$ 
  ((a - (-|a|) * b).toNat, -|a|, (Int.toNat_of_nonneg hnonneg).symm)
obtain ⟨q, hq⟩ := Nat.find_spec h
let r :  $\mathbb{N} :=$  Nat.find h
have hr0 : (0 :  $\mathbb{Z}$ )  $\leq$  r := Int.natCast_nonneg _

```

By minimality, for $s \in \mathbb{Z}$,

$$0 \leq s, \quad s = a - xb \ (x \in \mathbb{Z}) \implies r \leq s.$$

```

have hminimal :  $\forall s : \mathbb{Z}, 0 \leq s \rightarrow (\exists x : \mathbb{Z}, a - x * b = s) \rightarrow (r : \mathbb{Z}) \leq s$ 
   $\hookrightarrow$  := by
    intro s hs ⟨x, hx⟩
    have hn : Nat.find h  $\leq$  s.toNat :=
      Nat.find_min' h ⟨x, hx.trans (Int.toNat_of_nonneg hs).symm⟩
    have hc : (Nat.find h :  $\mathbb{Z}$ )  $\leq$  (s.toNat :  $\mathbb{Z}$ ) := Int.ofNat_le.mpr hn
    rwa [Int.toNat_of_nonneg hs] at hc

```

If $r \geq b$, then

$$0 \leq r - b = a - (q + 1)b < r,$$

contradicting minimality. Hence $r < b$.

```

have hrb_int : (r :  $\mathbb{Z}$ ) < b := by
  by_contra hnot
  have hs :  $0 \leq (r : \mathbb{Z}) - b :=$  by linarith
  have hsmaller := hminimal ((r :  $\mathbb{Z}$ ) - b) hs ⟨q + 1, by dsimp [r];
     $\hookrightarrow$  nlinarith [hq]⟩
  linarith

```

Thus $a = qb + r$ and $0 \leq r < b$, proving existence.

```

have ha : a = q * b + r := by dsimp [r]; linarith [hq]
have hrb : r < (b :  $\mathbb{N}$ ) := by exact_mod_cast hrb_int
refine ⟨(q, r), ⟨ha, hrb⟩, ?_⟩

```

Suppose another pair $(q', r') \in \mathbb{Z} \times \mathbb{N}$ satisfies $a = q'b + r'$ and $0 \leq r' < b$.

```
rintro (q', r') (ha', hr'b)
change a = q' * b + r' at ha'
have hr'0 : (0 : ℤ) ≤ r' := Int.natCast_nonneg _
have hr'b_int : (r' : ℤ) < b := by exact_mod_cast hr'b
```

Since $q, q' \in \mathbb{Z}$,

$$\begin{aligned} q' < q &\implies q - q' \geq 1 \implies r' = r + (q - q')b \geq b, \\ q' > q &\implies q' - q \geq 1 \implies r = r' + (q' - q)b \geq b. \end{aligned}$$

Both contradict the remainder bounds, so $q' = q$.

```
have hqq : q' = q := by
  rcases lt_trichotomy q' q with hlt | heq | hgt
  · have hstep : q' + 1 ≤ q := hlt
    nlinarith
  · exact heq
  · have hstep : q + 1 ≤ q' := hgt
    nlinarith
```

Then $r' = a - q'b = a - qb = r$, so $(q', r') = (q, r)$, proving uniqueness.

```
have hrr_int : (r' : ℤ) = r := by rw [hqq] at ha'; linarith
have hrr : r' = r := by exact_mod_cast hrr_int
exact Prod.ext hqq hrr
```

□

2.3 The Greatest Common Divisor

2.4 The Euclidean Algorithm

2.5 The Diophantine Equation $ax + by = c$

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Primes and Their Distribution

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16.2 Primality Testing and Factorization

16.3 An Application to Factoring: Remote Coin Flipping

16.4 The Prime Number Theorem and Zeta Function

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